

Milestone 2: Feature Engineering & Baseline Modeling

1. Project Overview

The goal of this milestone was to bridge the gap between raw pixel data and forensic identification. By extracting hand-crafted features, I aimed to identify the "digital fingerprints" (sensor noise and mechanical artifacts) unique to each scanner brand (HP, Canon9000-1, EpsonV550).

2. Feature Engineering

I focused on three domains of feature extraction to capture the physical characteristics of the scanners:

A. Noise Map Extraction (Spatial Domain): Because scanners have unique sensor imperfections (PRNU), I extracted the Residual Noise by applying a median filter and subtracting the denoised version from the original.

B. FFT Spectrum Analysis (Frequency Domain): I transformed the images using Fast Fourier Transform (FFT) to see periodic mechanical artifacts.

C. Texture Descriptors (LBP): Using Local Binary Patterns (LBP), I analyzed the local texture of the scans.

3. Baseline Modeling

To establish a performance benchmark, I trained two classical Machine Learning models. This allowed to quantify the difficulty of the problem before moving to Deep Learning.

A. Support Vector Machine (SVM)

- Performance: Achieved ~ 99.1%. It struggled to distinguish between Canon and Epson due to overlapping intensity distributions.

- SVM Accuracy: 0.9916666666666667
- Confusion Matrix:
- | | | |
|----|----|----|
| 39 | 1 | 0 |
| 0 | 40 | 0 |
| 0 | 0 | 40 |
- | | | |
|---|---|----|
| 0 | 0 | 40 |
|---|---|----|

B. Random Forest (RF)

- Performance: Achieved 99%.
 - Random Forest Accuracy: 0.9916666666666667
 - Confusion Matrix:
 - | | | |
|----|----|----|
| 39 | 1 | 0 |
| 0 | 40 | 0 |
| 0 | 0 | 40 |
 - | | | |
|---|---|----|
| 0 | 0 | 40 |
|---|---|----|
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4. Visual Evidence

- PCA Plot: Showed that the scanners' raw data clusters overlap significantly, proving the problem is non-linearly separable.
 - Confusion Matrix: Highlighting the "Confusion Zones" where the baseline models failed to tell the difference between similar sensor types.
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